

## 6.7.1 – 6.7.4 Spreadsheets, Presentations, Infographics, Reports

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے؛ اس کے لیے کینوا (Canva)...

CS XI

Chapter 6: Digital Literacy

Lecture 09 of 14

### 6.7.1 – 6.7.4 Spreadsheets, Presentations, Infographics, Reports

#### 6.7.3 Infographics

##### LEARN IT

An **infographic** (information + graphic) is a **visual way** of presenting information, facts or data so that people can understand the main idea **quickly and easily**. Tools such as **Canva**, **Power BI** or **Tableau** can be used. A good infographic should:

- **Simplify** complex concepts.
- **Highlight** the most important findings and results.
- Use **clear visuals**, **limited text** and **accurate** information.

#### 6.7.4 Reports

##### LEARN IT

**Reports** are **structured documents** used to present collected data, findings, analysis and conclusions. Tools like **MS Word** or **Google Docs** are standard. A good report should:

- Start with a clear **title**, **introduction** and **purpose**.
- Present the collected data using **tables or charts**.
- Explain findings in **simple language**.
- Include **conclusions and recommendations** based on the data.

**Optional:** you may use AI tools carefully to improve wording, but **always review for accuracy**. AI can make mistakes or invent facts, and you are responsible for what you submit.

##### LEARN IT

**Visual rule of thumb:** a good chart or infographic can explain complex data **faster** than a long paragraph. However, visuals must always be **accurate** and based strictly on the **real data collected** – never exaggerated.

### WORKED EXAMPLE

#### Which tool should you choose?

| Situation                                                           | Best tool    | Why                                 |
|---------------------------------------------------------------------|--------------|-------------------------------------|
| You must calculate the average marks of 40 students.                | Spreadsheet  | Formulas calculate instantly.       |
| You will explain results to the class in assembly.                  | Presentation | Designed for a live audience.       |
| You want a one-page poster for the notice board.                    | Infographic  | Main result understood at a glance. |
| The principal wants the full study with method and recommendations. | Report       | Detailed, structured document.      |

### REMEMBER IT

Match tool to job: **Spreadsheet = Calculate**, **Presentation = Tell**, **Infographic = Show quickly**, **Report = Explain fully**.

### COMMON MISTAKE

**Mistake:** Making a chart look more dramatic than the data – for example, starting the bar-chart axis at 18 instead of 0 so that 20 looks twice as big as 19. **Fix:** Start bar-chart axes at 0, label axes, and show real numbers only.

### اردو میں سمجھیں

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے؛ اس کے لیے کینوا (Canva) اور پاور بی آئی (Power BI) جیسے ٹولز استعمال ہوتے ہیں۔ رپورٹ ایک منظم دستاویز ہے جس میں عنوان، تعارف، ڈیٹا، نتائج، اختتامیہ اور سفارشات شامل ہوتی ہیں۔ تصاویر اور چارٹس ہمیشہ اصل ڈیٹا کے مطابق اور درست ہونے چاہئیں، مبالغہ آمیز نہیں۔

### CHECK YOURSELF

1. Name two tools for making infographics.
2. What must you always do if you use AI to improve your report's wording?
3. What should a report end with?

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Answers: 1. Any two: Canva, Power BI, Tableau. 2. Review it carefully for accuracy. 3. Conclusions and recommendations based on the data.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Spreadsheets are good for:

- a. Tables, calculations, simple charts
- b. 3D games
- c. NAND simulation
- d. OSI sniffing

**MCQ 2.** A presentation (slides) is for:

- a. Talking an audience through findings
- b. Storing the only copy of raw data
- c. Primary keys
- d. Kernel drivers

**MCQ 3.** An infographic is:

- a. A visual one-pager of facts
- b. A 40-page appendix
- c. A binary search
- d. A foreign key

**MCQ 4.** A report is:

- a. Structured writing with method, results, limits
- b. Only memes
- c. Only code
- d. Only ER

**MCQ 5.** Choose the tool by:

- a. Audience and purpose
- b. Whatever is newest
- c. Always infographic
- d. Always 80 slides

**MCQ 6.** Raw 2000-row sheets in a 5-minute talk are:

- a. A bad present
- b. Ideal
- c. Required by BIEK
- d. A sort algorithm

## Extra short questions

1. Four digital presentation tools in the book.
2. When spreadsheet > slides.
3. When infographic > report.
4. One rule for all four.

## Extra long question

1. Same finding (70% of surveyed students want longer labs). Design: a spreadsheet cell, a slide headline, an infographic hook, and a report paragraph.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Spreadsheets, slides, infographics, reports.
2. When the audience must check the numbers or filter.
3. Awareness poster / social post — not a methods thesis.
4. Cite sources; label axes; do not crop axes to lie.

**Long**

1. Sheet: table+bar. Slide: one number, one chart, one ask. Infographic: 7 of 10 icon row.  
Report: sample n, question wording, 70%, limit (one college).

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Next lecture: 6.8 Case Study: Digital Inquiry Project